

[illegible]

INTRODUCTION TO BIOLOGY
BIO-101

Evolution and Diversity of Life
Lecture 01

What does Biology mean?

Biology Comes from two Greek words:

- “Bios” - means life
- “Logos” - means study
- Study of life, i.e. the study of living things at many scales: from individual cells to multicellular organisms to entire biosphere!
- Their environment and the interaction between them.

Importance of Biology

Enables us to understand:

- “Life”
- Evolution & diversity
- Causes of diseases, leading to discovery of new medications
- How to preserve nature
- Develop technologies to improve life style: medical, environmental, agricultural

In a whole, **it helps you develop, modify, and refine your ideas about life and thus ensures in creating a healthy biosphere.**

**Field of study
in biology**

virology
Study of
viruses

anatomy
Study of org.
structure

Genetic
Study of
heredity

entomology
Study of
insects

ecology
Study of envi.
and
relationship

zoology
Study of
animal

microbiology
Study of
microorg.

embryology
Study of
embryonic dev.

biochemistry
Study of
biochemical
process in body

botany
Study of
plant

bacteriology
Study of
bacteria

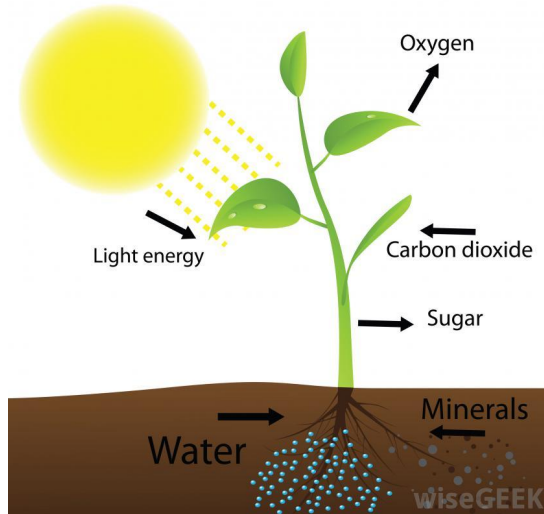
taxonomy
Study of
classifying
plants &
animals

mycology
Study of
fungi

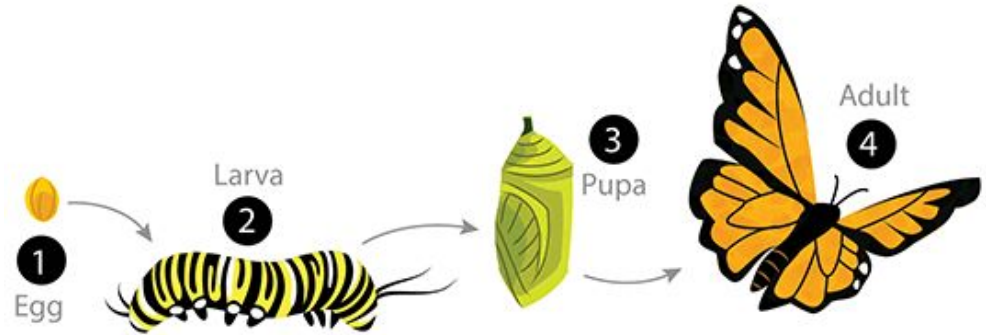
histology
Study of
tissues

What Is Life?

- Life is something that **captures and uses energy and raw materials**, this energy is the capacity to do work.



- Living organisms have the capacity to do metabolism and this is a set of reactions where cells acquire and use energy to **grow, survive, reproduce and develop**.



Characteristics of Life:

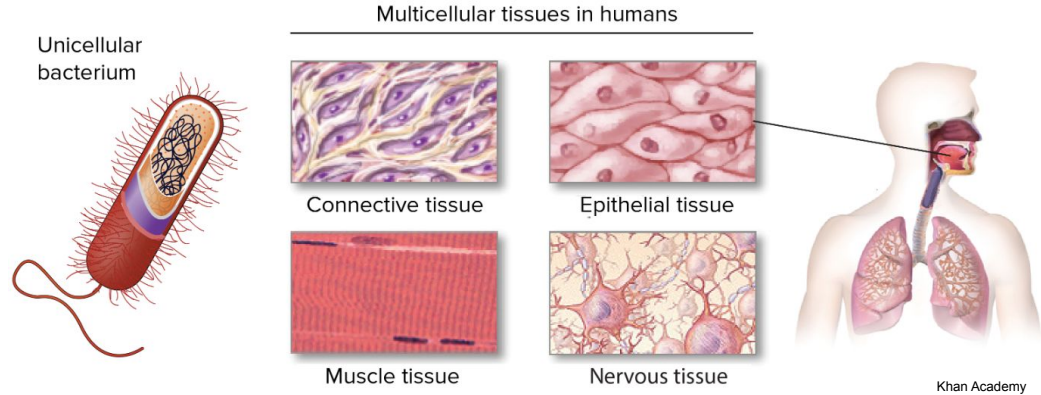
1. Made of one (unicellular) or more cells (multicellular)

Unicellular

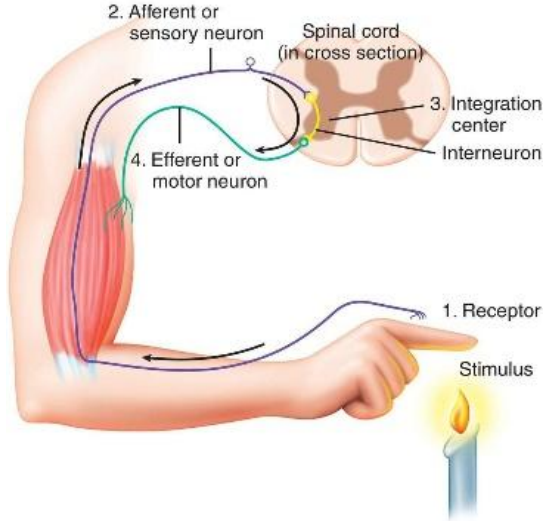
- made up of a single cell
- single cell performs all tasks
- shorter life span

Multicellular

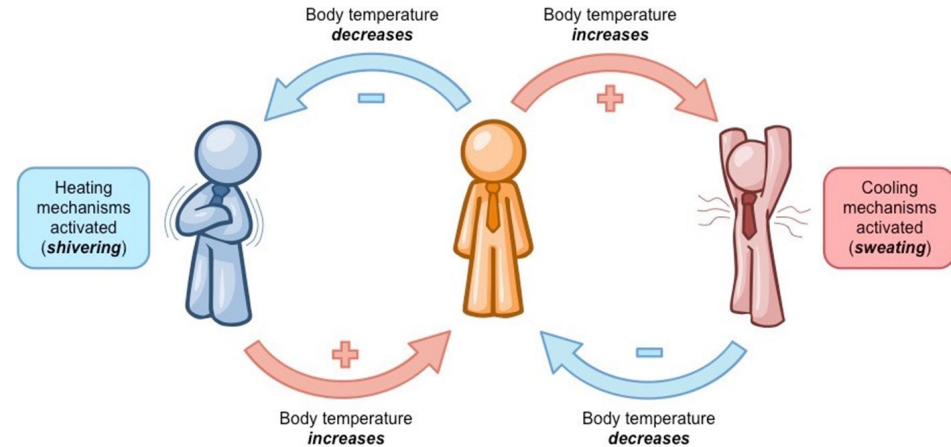
- made up of a multiple cells
- many cells work together, each has their own function
- longer life span, damaged cells can be replaced



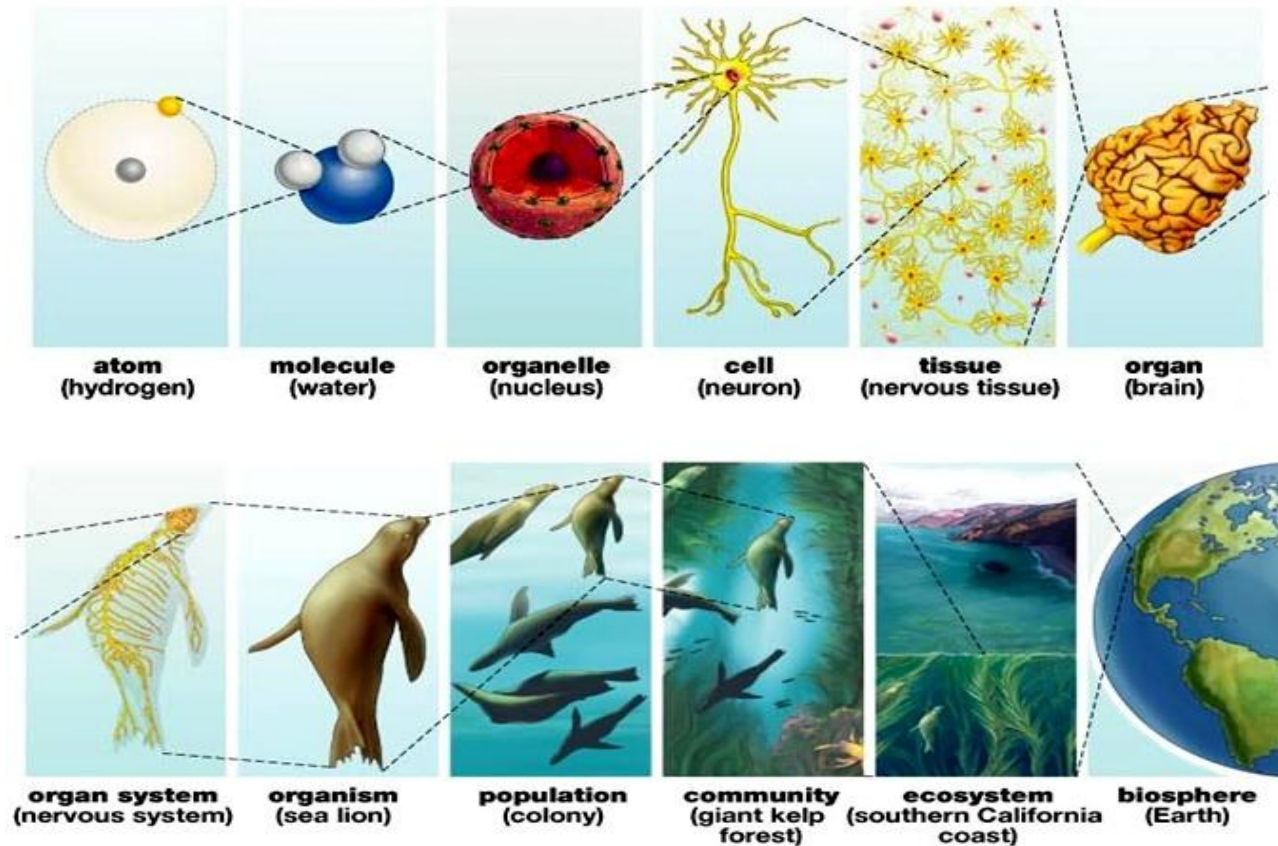
2. Living organisms can **sense changes in the environment and respond according to the changes**. Organisms have receptors that detect specific forms of energies which then are called stimuli (Singular stimulus).



3. Maintains **homeostasis that is the maintenance of internal environment within range suitable for cell activities**. Example: Maintenance of blood sugar level by pancreas or balancing the temperature of the body.



4. Display Higher level of organization



5. Grow, Develop and Reproduce

Growth: Adding mass

Development: Natural changes over the lifetime of an organism

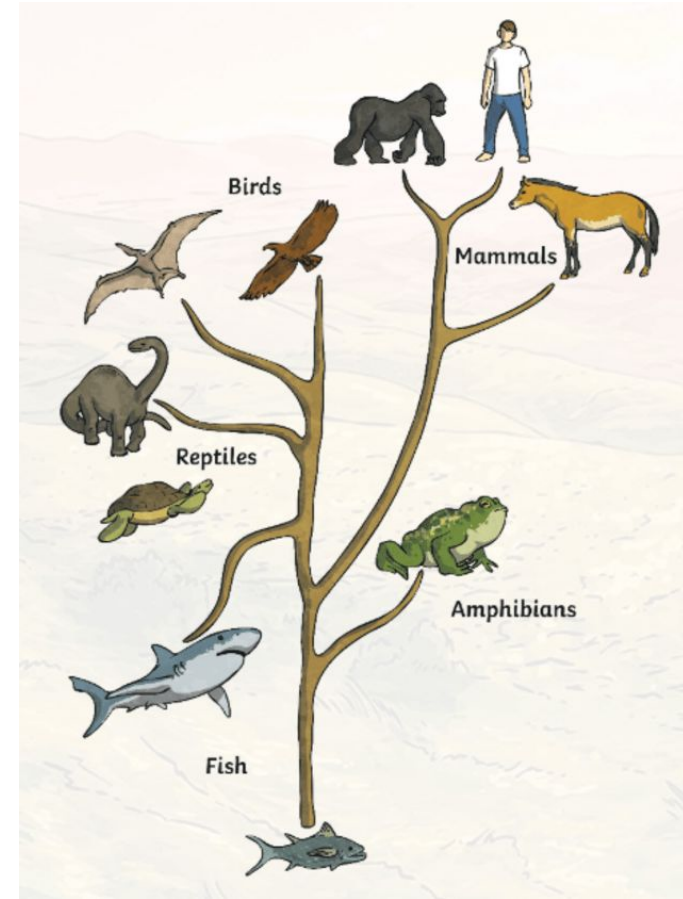
Reproduction: Production of new offspring - essential for the continuation of life.

6. Life has the tendency to evolve which we call evolution.

Evolution is the genetically based change in population over time.

Life adapts with time and the organisms best adapted to the environment are more likely to survive and reproduce.

This is known as _____??



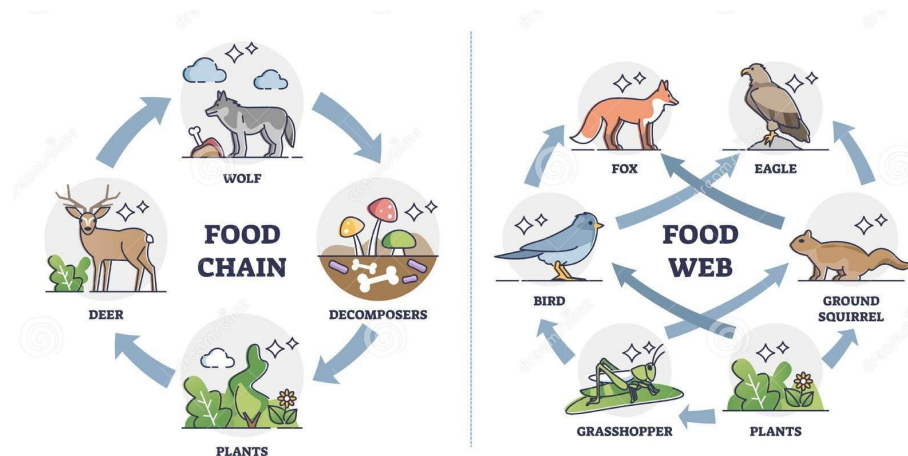
Ecosystem

- Living organisms + physical environment (non living)
 - Communities of organisms living together in combination with their physical environment.
 - Living organisms of **different species** in an ecosystem is called a **community**
 - A **population** is only **one species** (one type) of organism.
- A food web is a pattern that shows the transfer of energy between different organisms (producers and consumers). Food chain??

All energy in a food web comes from sun, through ____?

The three members in every community-

- Producers- can make their own food
- Consumers- feed on producers/other consumers
- Decomposers-break down wastes and remains of other organisms



Interdependencies Among Organisms

Producers

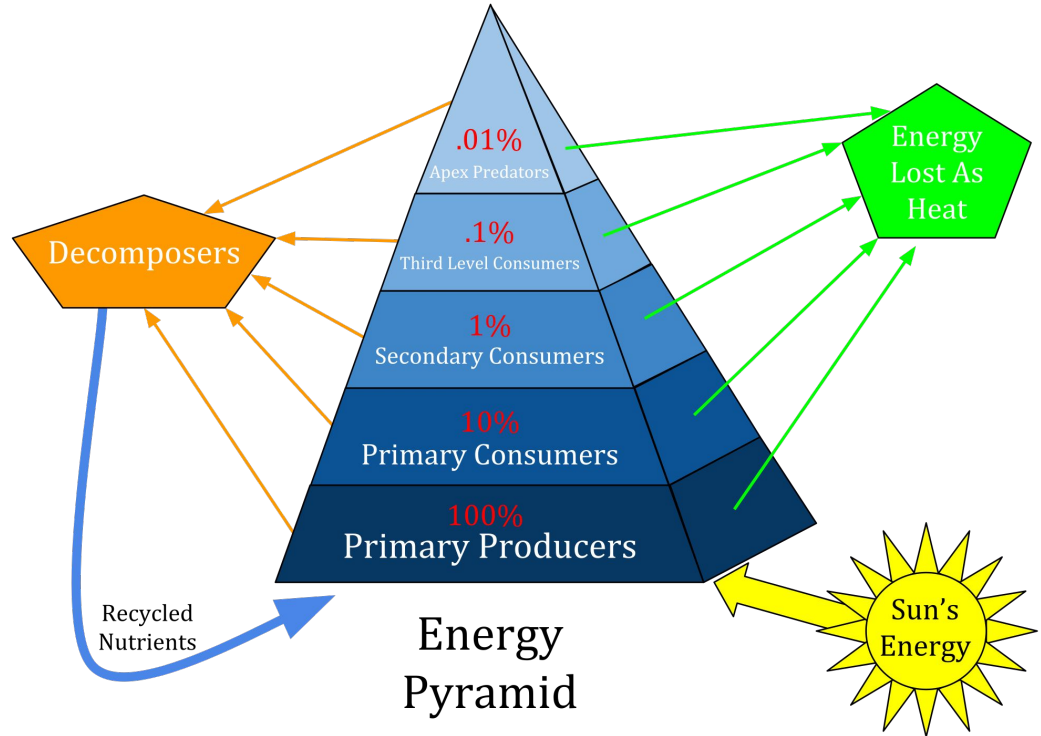
Make their own food

Consumers

Depend on energy stored in tissues of producers

Decomposers

Break down remains and wastes



Energy Flow

- Usually starts with energy from sun
- Transfer from one organism to another
- Energy flows in one direction
- Eventually, all energy flows back to the environment

The Flow of Energy Through Ecosystems

Producers, Consumers, and Decomposers

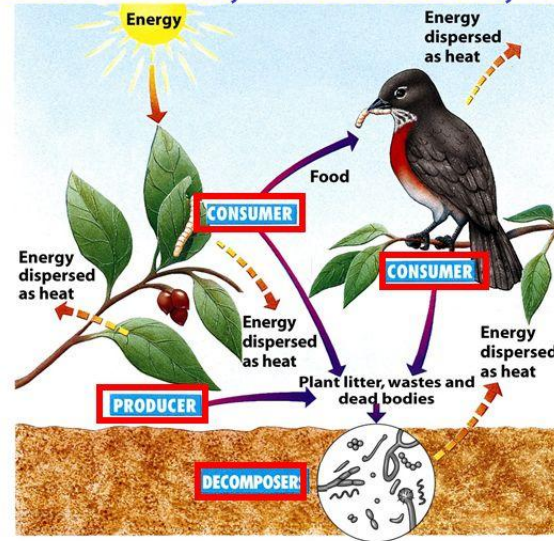


Figure 3-8b Environment, 5/e
© 2006 John Wiley & Sons

Energy flows
from Producers

To Consumers

And finally to
Decomposers

- A habitat is the home of an organism.
- The size of the habitat depends on the organism's needs.
- An organism's habitat provides food, water, shelter, and space.
- The ecological role of an organism in its habitat is called niche.
- Usually two types of organisms do not occupy exactly the same niche in a community.
- During its life cycle, an organism's niche may change.



Evolution

Evolution is defined as the changes in a species over time.

- Much of theories we know about the history of life on earth, comes from fossils.
- The fossil record --> life forms have gradually changed over time.

Natural Selection

- “Natural selection is the process through which populations of living organisms adapt and change.” - National Geographic
- The process of natural selection causes organisms to evolve!

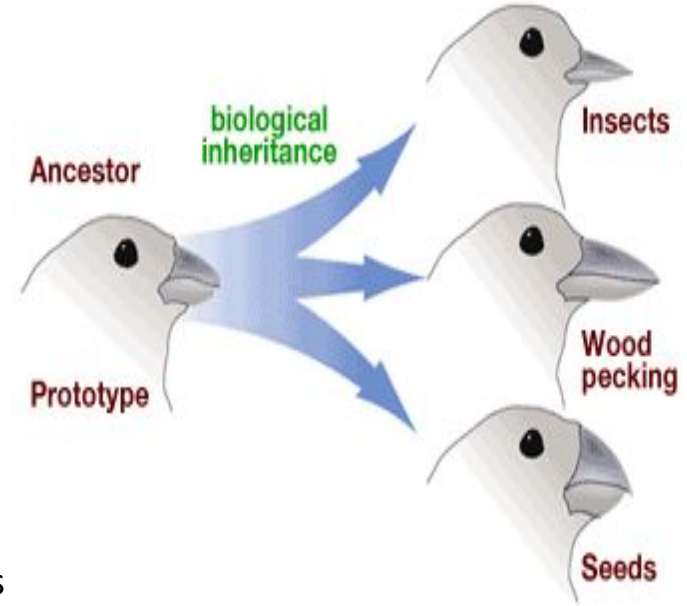
Organisms change over time

Adaptive trait

A trait that gives the individual an advantage in survival or reproduction, under a given set of circumstances

For example, in finches, the beak shape can be an adaptive trait. Species that eat large seeds tends to have large, tough beaks, while those that eat insects had thin, sharp beaks.

- Baby deer: light brown fur and white spots to hide on leafy forest floors
- Chameleons- **special skin cells** that allows them to change colors
- Walking stick insects looking like tree twigs



Antibiotic Resistance!

Another common example of Natural selection

- Antibiotics are used to kill bacteria
- Mutations for antibiotic resistance exist or arise in bacterial population
- Antibiotic-resistant bacteria survive and reproduce better than non-resistant
- Over time, proportion of antibiotic-resistant bacteria increases

How does antibiotic resistance occur?



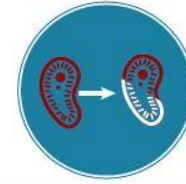
1 Some bacteria in the human body are drug resistant.



2 Antibiotics kill bacteria, but not those resistant to drugs.



3 Resistant bacteria then have space to multiply.

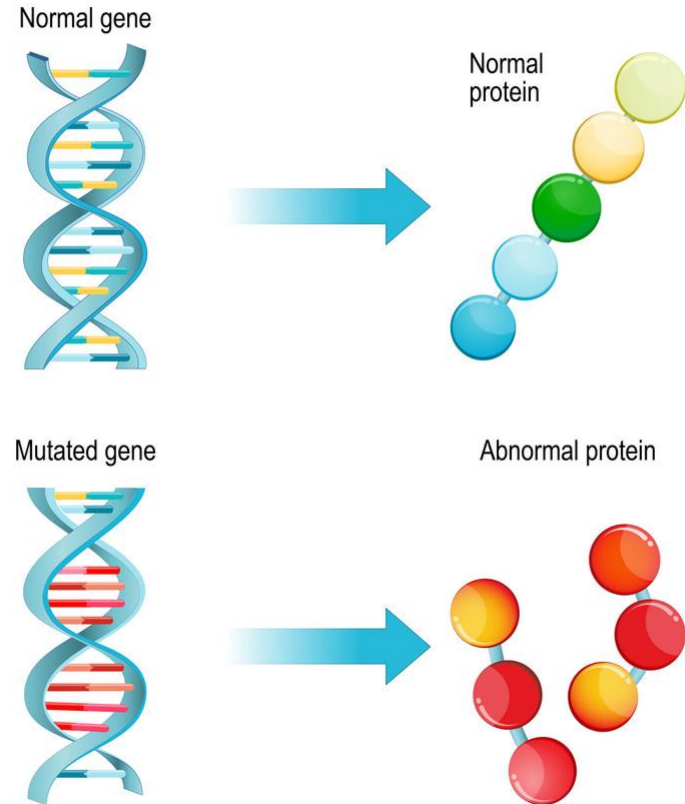


4 Bacteria can even transfer their drug resistance to other bacteria.

Mutation

- Changes in the predicted genetic pattern (DNA sequence) of organisms.
- They can be caused:
 - randomly
 - exposure to mutagens: agents that can induce mutation
- Mutagen examples: X-ray, UV, wide variety of chemicals and even viruses
- Sometimes scientists use mutations for their advantages.
Examples are breeding livestock/ crops with desirable traits: seedless grapes!

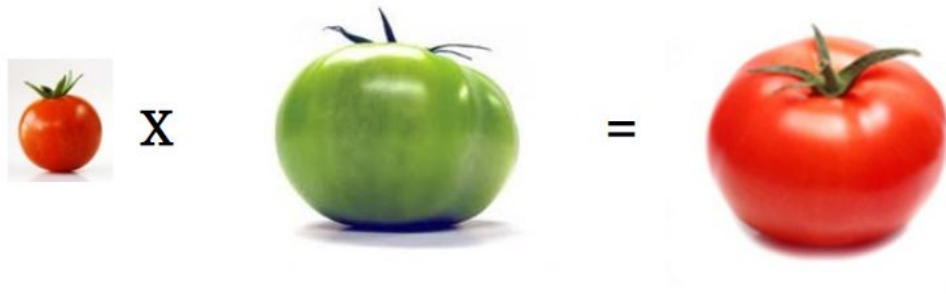
Genetic mutation



Artificial Selection

Selective breeding of organisms to promote the appearance of desirable traits in offspring is called artificial selection.

- Breeders favor some form of traits over others
- Individuals exhibiting favored traits are bred
- Favored traits increase in the population



Classification of Life

The three-domain system is a biological classification introduced by Carl Woese, Otto Kandler, and Mark Wheelis in 1990 that divides cellular life forms into three domains, namely Archaea, Bacteria, and Eukarya.

Domain Archaea: The Archaea are prokaryotic (lacks nucleus). Oldest species on Earth. Most notable: their diverse, exotic metabolisms. Examples:

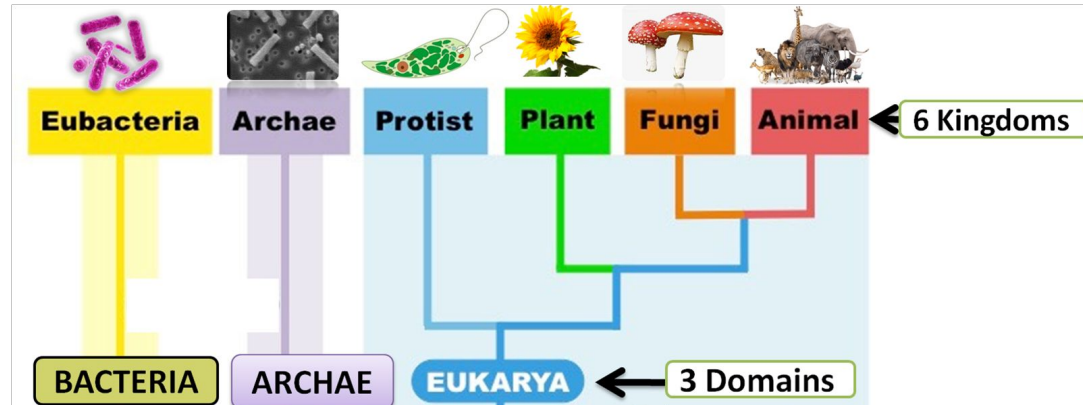
- methanogens – which produce the gas methane
- halophiles – which live in very salty water
- thermoacidophiles – which thrive in acidic high-temperature water

Domain Bacteria

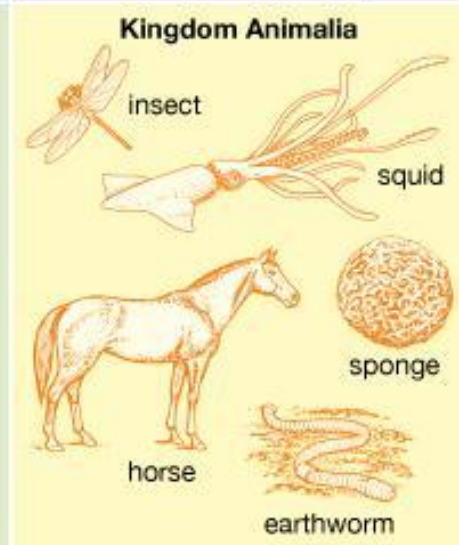
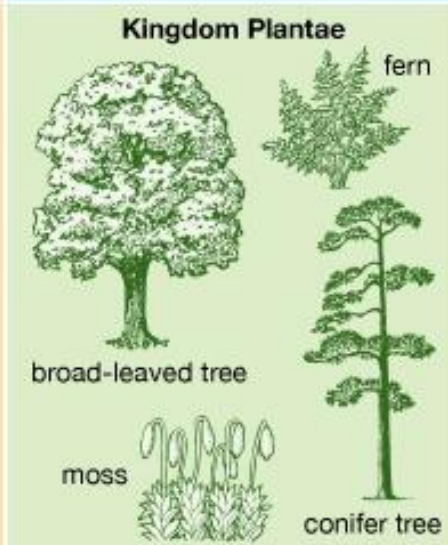
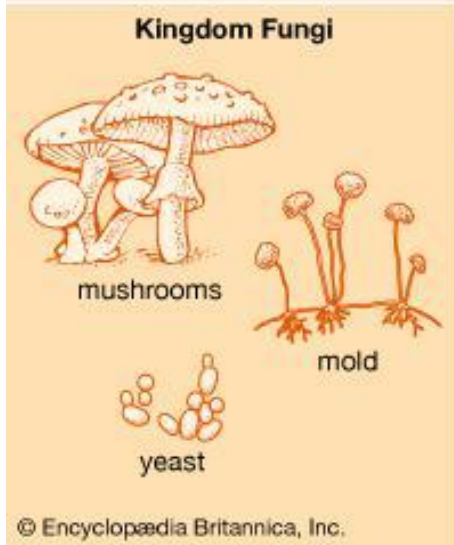
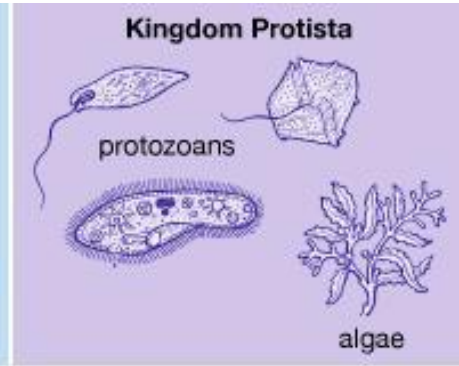
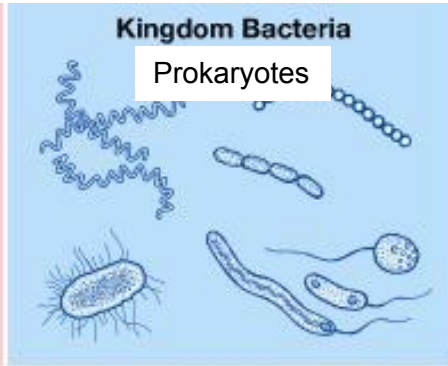
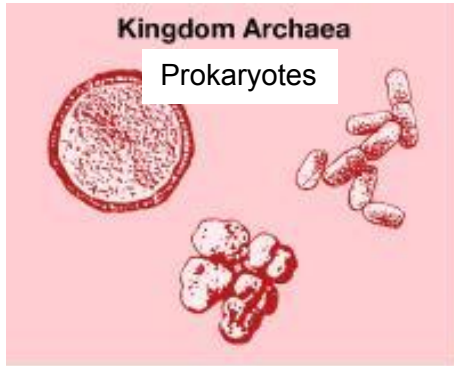
The Bacteria are also prokaryotic; their domain consists of cells with bacterial genetic material and lacks nucleus.

Domain Eukarya

Eukaryotes are organisms whose cells contain a membrane-bound nucleus. They include many single celled organisms to all multicellular organisms.



The Six Kingdoms



Protistans

- Not an easily defined group
- Producers and consumers
- Single-celled and multicellular species



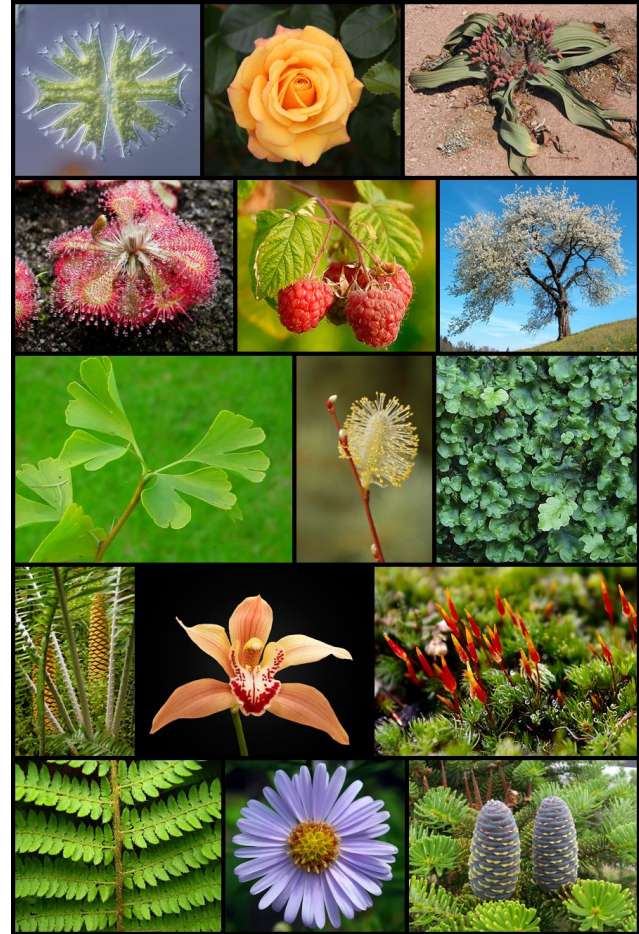
Fungi

- Most are multi celled
- Consumers and decomposers
- Extracellular digestion and absorption



Plants

- All are multi celled
- Most are photosynthetic producers
- Make up the food base for communities, especially on land



Animals

Multi Celled consumers

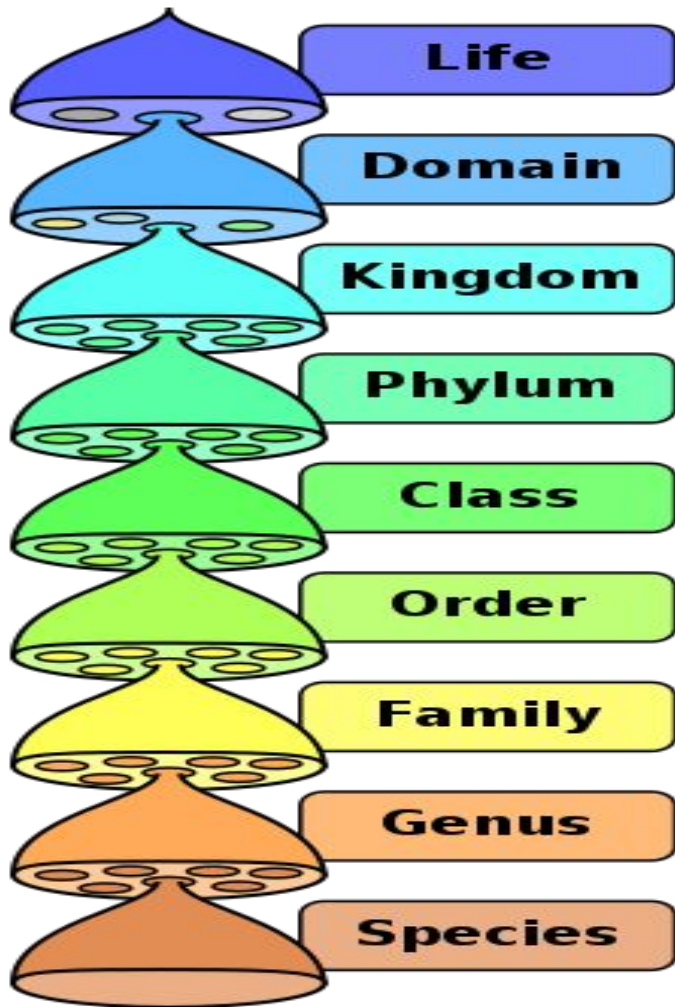
- Herbivores
- Carnivores
- Parasites
- Scavengers

Move about during at least
some stage of their life



Taxonomy

- Science that deals with naming, describing and classifying all living organisms
 - Classifying organisms is based on similarities between different organisms.
 - A biologist named Carolus Linnaeus started this naming system. He also chose to use Latin words.
-
- This naming system have different parts. They describe an organism in a series of general names to specific names.



Phylum

Phylum is the primary division of kingdom. It includes one or more related classes of animals

Class

Class is consisting of one or more related orders.

Order

Order contains one or more families.

Family

Family contains one or more related genera.

Genus

Genus includes closely related species.

Species

Very specific!

Taxonomy



Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
Order:	Carnivora
Suborder:	Feliformia
Family:	Felidae
Subfamily:	Pantherinae
Genus:	<i>Panthera</i>
Species:	<i>P. tigris</i>

Binomial name

Panthera tigris
(Linnaeus, 1758)^[2]



Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
Order:	Carnivora
Suborder:	Feliformia
Family:	Felidae
Subfamily:	Felinae
Genus:	<i>Felis</i>
Species:	<i>F. catus</i> ^[1]

Binomial name

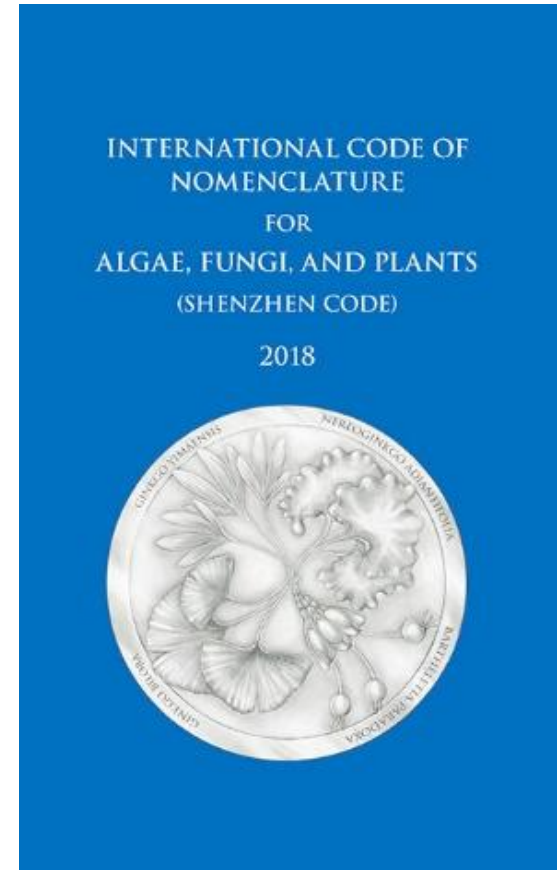
Felis catus^[1]
Linnaeus, 1758^[2]

Scientific Names

Every known living organism on Earth is classified and named by a set of rules. The names are called **scientific** names. It is a formal naming system, widely used.

Common names are the ones you might use when talking with your friends. You call your pet, a dog (the common name) and not *Canis familiaris*.

Vary, depending on location.



Scientific Names (cont.)

Scientific names follow a specific set of rules. Scientists use a two-name system called a **Binomial Naming System**. Scientists name animals and plants using the system that describes the genus and species of the organism. A binomial name means that it's made up of two words (bi-nomial).

- **The first word is the genus and the second is the species.**
- **The first word (genus) is capitalized and the second (species) is not.**

Humans are scientifically named *Homo sapiens*. You may also see an abbreviation of this name as *H. sapiens* where the genus is only represented by the first letter.

- **In first use, the genus is always written in full. In subsequent uses, the genus can be abbreviated: first initial followed by a period.**

- **When you type: italicize!**

Binomial Nomenclature



Common Name: Grizzly Bear
Scientific Name: *Ursus arctos*

Rule for binomial nomenclature

First Name **Genus** and is always **Capitalized**
Second Name **Species** and is **not capitalized**



Common Name: Pebble Crab

Genus Species
↓ ↓
Scientific Name: *Xanthias* *lamarckii*

